

Project: Courier & Parcel Management System

Technology Stack

- Framework: Laravel 12
- PHP: 8.3+
- Database: MySQL
- Frontend: Blade + Bootstrap/Tailwind CSS (*Vue.js or React will be considered an advantage*)
- Authentication: Custom Login & Registration

Objective

Develop a Courier & Parcel Management System that enables administrators to efficiently manage customers, drivers, branches, parcels, deliveries, and shipment tracking.

The project should demonstrate your knowledge of Laravel development, clean architecture, database relationships, authentication, authorization, RESTful APIs, validation, and application optimization.

Required Modules

1. Authentication

Implement a secure authentication system.

Features

- User Registration
- User Login
- User Logout
- Forgot Password
- Change Password
- Email Verification (*Bonus*)

2. User Management

Implement role-based user management.

User Roles

- Super Admin
- Branch Manager
- Dispatcher
- Driver

Features

- Create Users
- Update User Details

- Delete Users
- Assign User Roles
- Activate/Deactivate User Accounts

3. Customer Management

Create a customer management module with complete customer information.

Customer Information

- Customer ID (Auto Generated)
- Full Name
- NIC / Passport Number
- Mobile Number
- Email Address
- Address
- City
- Postal Code
- Company Name (*Optional*)
- Status

Features

- Complete CRUD Operations
- Search Customers
- Filter Customers
- Pagination
- Soft Delete
- Customer Profile Page
- View Customer Shipment History

4. Driver Management

Manage delivery drivers and their information.

Driver Information

- Driver Code
- Full Name
- Phone Number
- Email Address
- Vehicle Number
- License Number
- Vehicle Type
- Assigned Branch
- Status

Features

- Complete CRUD Operations
- Upload Driver Profile Photo
- Activate/Deactivate Driver
- View Assigned Deliveries
- Driver Performance Summary

5. Branch Management

Manage courier branches.

Branch Information

- Branch Name
- Address
- Contact Number
- Branch Manager

Features

- CRUD Operations
- Assign Drivers to Branches
- View Branch Shipments

6. Parcel Management

Manage parcel registration and delivery information.

Parcel Information

- Tracking Number (*Auto Generated*)
- Customer
- Receiver Name
- Receiver Phone Number
- Receiver Address
- Pickup Address
- Parcel Type
- Weight
- Dimensions
- Delivery Charge
- Payment Method
- Payment Status
- Delivery Priority

Delivery Priority

- Normal
- Express
- Same Day

Parcel Status

- Pending
- Picked Up
- At Warehouse
- Out for Delivery
- Delivered
- Failed Delivery
- Returned

Features

- Create Parcel
- Update Parcel Details
- Cancel Parcel

- Print Shipping Label
- Upload Parcel Images

7. Shipment Tracking

Every parcel should maintain a complete tracking history.

Example Tracking Flow

- Package Created
- Picked Up
- Received at Warehouse
- Sorted
- Dispatched
- Out for Delivery
- Delivered

Tracking History

Each tracking record should include:

- Parcel
- Status
- Location
- Updated By
- Date & Time
- Remarks

Feature

- Customers should be able to search parcel status using the **Tracking Number**.

8. Delivery Management

Manage parcel delivery assignments and completion.

Features

- Assign Parcels to Drivers

Driver Actions

- Accept Delivery
- Reject Delivery
- Complete Delivery
- Update Delivery Status

Delivery Details

- Delivery Time
- Delivery Location
- Receiver Name
- Receiver Signature (*Bonus*)
- Delivery Image (*Bonus*)

9. Dashboard

Create an informative dashboard displaying key business statistics.

Dashboard Widgets

- Today's Shipments

- Today's Deliveries
- Pending Deliveries
- Delivered Parcels
- Failed Deliveries
- Total Revenue
- Available Drivers
- Drivers on Delivery
- Top Customers
- Top Performing Drivers

Charts

- Monthly Shipments
- Monthly Revenue
- Delivery Success Rate

10. Reports

Generate business reports.

Reports

- Daily Shipment Report
- Monthly Revenue Report
- Driver Performance Report
- Customer Shipment Report
- Delivery Report

Export Options

- CSV
- Excel
- PDF (*Bonus*)

11. API & Development Requirements

The project should follow Laravel best practices.

Requirements

- Laravel Sanctum Authentication
- Form Request Validation
- API Resources
- RESTful API Structure
- Proper HTTP Status Codes
- Pagination
- Clean Folder Structure
- Database Relationships
- Eloquent ORM Best Practices

12. Search & Filtering

Implement search and filtering functionality.

Customers

- Name

- Mobile Number
- Email Address

Drivers

- Branch
- Status

Parcels

- Tracking Number
- Status
- Customer
- Date Range
- Assigned Driver
- Branch

13. Validation Rules

Implement appropriate server-side validation.

Examples

- Tracking Number must be unique.
- Parcel Weight must be numeric and greater than zero.
- Phone Number must follow the Sri Lankan mobile number format.
- Email Address must be valid and unique where applicable.

Bonus Features

QR Code Generation

Generate a unique QR Code for every parcel.

Scanning the QR Code should display:

- Tracking Number
- Sender Details
- Receiver Details
- Current Shipment Status

Deliverables

Candidates should submit:

- Complete Laravel Project Source Code
- MySQL Database Export or Migration Files
- README File with Installation & Setup Instructions
- API Documentation (if APIs are implemented)
- Screenshots and a short demonstration video (*Required*)